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Sixteen years of two-stage floodplain development: Long-term changes in soil chemistry

Brian A. Wagner, Jennifer L. Tank, Mitchell J. Liddick *, Ursula H. Mahl

University of Notre Dame, Holy Cross Dr, Notre Dame, IN, USA

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ABSTRACT

High concentrations of nitrogen (N) and phosphorus (P) are often found in subsurface tile drain effluent from agricultural fields resulting in eutrophication and subsequent dead zones in downstream bodies of water. Although two-stage ditches are widely implemented as floodplain restoration practices to improve channel stability and enhance nutrient retention, their long-term (>10 years) soil biogeochemical development and capacity to function as sustained nutrient sinks remain poorly quantified. To evaluate the long-term nutrient retention potential of two-stage ditches, we assessed soil biogeochemical properties across a chronosequence of restored floodplain sections in the Shatto Ditch Watershed, IN, USA, spanning 5–16 years since construction. Additionally, we compared floodplain soils with both the on-floodplain flowpaths of tile drain effluent and adjacent agricultural fields across depth profiles, providing context to evaluate how soil properties changed relative to the fields from which the floodplains were constructed. We found that soil % organic matter (OM) on 5- and 6-year floodplains was comparable to their respective adjacent fields. In contrast, % OM on the 16-year floodplain was significantly higher than both its adjacent field and the younger floodplains, consistent with progressive OM enrichment over time. We found that soil N and P also suggested progressive accumulation with floodplain age, with N significantly higher in the 16-year floodplain relative to both its adjacent field and the younger floodplains, and P trending higher in older floodplains. Our results indicate that two-stage ditches accumulate OM, N, and P over 16 years, highlighting their potential as long-term nutrient retention features in agricultural landscapes.

1. Introduction

Agriculture significantly impacts water quality, with nutrient runoff causing eutrophication that can stimulate algal proliferation in downstream ecosystems (Goolsby et al., 1999; Rabalais et al., 1996; Vidon et al., 2012). Subsequent decomposition of large algal biomass quantities can create hypoxic “dead zones” in these downstream ecosystems (Conroy et al., 2014; Michalak et al., 2013), which have devastating effects on the fishing industry, natural biodiversity, and food web dynamics (Diaz, 2001). Harmful algal blooms and subsequent periods of hypoxia have increased in frequency and spatial distribution in recent decades (Diaz, 2001; Feng et al., 2024); as such, the reduction of nutrient runoff from agricultural landscapes has been a major focus area for conservation efforts (Mississippi River/Gulf of Mexico Watershed Nutrient Task Force, 2023 Report to Congress No. 4., 2023).

Low-gradient agricultural lands are frequently modified by artificial drainage networks of channelized ditches and subsurface tile drainage, which reduce standing water on fields, improving cash crop yields (Kim et al., 2022). Conventional channelized ditches are constructed in a trapezoidal cross-sectional morphology (Fig. 1A) which limits the ability of stream water to interact with riparian plants and soil microbiota during high flow events (Ducros and Joyce, 2003; Fennessy and Cronk, 1997). Therefore, there is limited potential for nutrient removal from agricultural runoff since the water bypasses grass buffer strips as it is rapidly shunted into the adjoining drainage networks by tile drains (de Wit et al., 2002; Vitousek et al., 1997). These subsurface drainage networks are a major source of nutrient contamination, exacerbating algal blooms from excess surface water and groundwater inputs. In particular, concentrations of dissolved nitrogen (N) and phosphorus (P) are often much higher in agricultural runoff than natural water sources because of excess fertilizer application to agricultural fields (Goolsby et al., 1999; Vidon et al., 2012).

Floodplain restoration via the construction of a two-stage ditch (Fig. 1A, Fig. 1B) is an in-stream practice that can be stacked with on-

* Corresponding author.
E-mail address: mliddick@nd.edu (M.J. Liddick).

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field approaches (e.g., winter cover crops) to improve nutrient retention and stream bank stability (Christopher et al., 2017; D'Ambrosio et al., 2015; Davis et al., 2015). The two-stage ditch is constructed by removing the top layers of soil on either side of the main channel to create floodplains (Fig. 1A, Fig. 1B) (Frankenberger et al., 2014). During storms or other high flow events, floodplains can become inundated, increasing the width of the wetted channel which reduces water velocity. This promotes sediment deposition, decreases bank erosion relative to trapezoidal ditches, and increases the surface area available for biogeochemical nutrient removal (Powell et al., 2007a, 2007b). By increasing bioreactive surface area (up to 3x) and decreasing water velocity, the two-stage ditch also increases space and time for biogeochemical processes that remove dissolved nutrients from the water column (Davis et al., 2015; Opdyke et al., 2006). At baseflow, stream flow is confined to the main channel and effluent from tile drains flows across floodplains prior to entering the stream in the two-stage ditch. At high flow, floodplains are inundated, providing increased interactions between the stream water and the physical, chemical, and biological components of the floodplain. During baseflow, two-stage ditches can also provide a buffer between tile drain outlets and the stream, promoting N and P removal along the flowpath between the tile drain and the stream (Fig. 1A).

The adoption of the two-stage management practice has been shown to improve downstream water quality. A study using the Soil Water Assessment Tool estimated that widespread implementation of two-stage in the River Raisin Watershed in the western Lake Erie Basin, USA could decrease annual nitrate (NO_3^--N) and total P export by 10 and 31%, respectively (Christopher et al., 2017). Inundation of two-stage floodplains promotes NO_3^--N removal via microbially-mediated denitrification, which is enhanced by increased bioreactive surface area, longer water residence times, and the formation of persistent anoxic microsites in floodplain soils (Briggs et al., 2015; Harvey et al., 2013; Opdyke et al., 2006; Roley et al., 2012b; Sorensen, 1978; Speir et al., 2020). Furthermore, previous studies have documented higher denitrification rates on older two-stage floodplains compared to newer ones, which was in turn linked to higher soil organic matter (OM) on older floodplains (Mahl et al., 2015; Speir et al., 2020). These results suggest that two-stage function can improve over time through the establishment of microbial communities and vegetation after construction, and the natural cycle of plant growth, senescence, death, and decomposition that occurs annually on floodplains, increasing carbon (C) availability and promoting denitrification (Burchell et al., 2007; Zhai et al., 2013).

While the effects of the two-stage on P are less well-studied, previous research shows that two-stage ditches can remove soluble reactive P from stream water via adsorption to floodplain soils and uptake by vegetation (Davis et al., 2015; Trentman et al., 2020). Additionally, sediment deposition on floodplains can reduce total P by removing particulate P from the water column (Grayson et al., 1996). However, given that the anoxic conditions that promote denitrification can also promote P release from soils, there is concern that two-stage construction may increase P export by exposing legacy P in agricultural soils to the stream during floodplain inundation (Davis et al., 2015; Torbert et al., 2002).

In addition to enhancing denitrification, accumulation of OM on two-stage floodplains could influence P retention due to pH-dependent complexation between P and soil cations such as aluminum (Al), iron (Fe), and calcium (Ca) (Penn and Camberato, 2019). Soil Al exists in many different chemical complexes (Aide, 2022) and its toxicity to plants has been extensively documented (Barceló and Poschenrieder, 2002; Delhaize and Ryan, 1995; Sivaguru and Horst, 1998). In a study of Ultisol and Alfisol soils, OM content was shown to be negatively correlated with soluble soil Al, while lower soil pH was shown to increase soil Al activation (Li et al., 2022), thereby permitting Al-P complexation. Conversely, another study showed that an increase in OM led to an

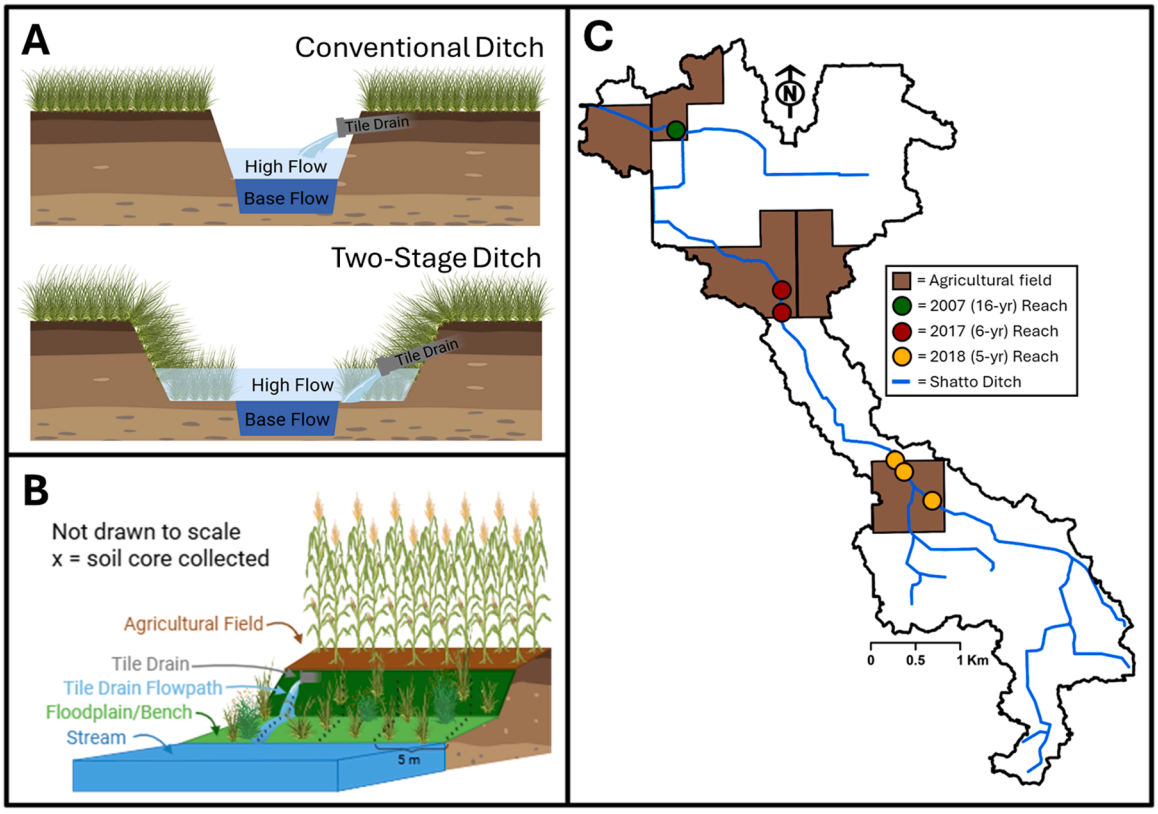


Fig. 1. Two-stage ditch diagram and map of Shatto Ditch Watershed study sites. A: Cross-section of a conventional channelized ditch (top) versus a two-stage ditch (bottom). B: We collected 10 soil cores (each represented by a black "x" and separated by 0.5 m) along three transects perpendicular to stream flow (5 m apart), in addition to the tile drain flowpath. C: Map of SDW in Kosciusko County in north-central Indiana, USA depicting study sites.

increase in Fe as measured by the Mehlich III soil test; however, the additional Fe occurred as organic and exchangeable forms, not soluble forms, meaning plants could uptake it but it would not be immediately available to complex P (Shuman, 1988). Finally, Ca-mediated stabilization of soil OM has led to findings of positive correlations between Ca and soil OM; different mechanisms account for OM stabilization, including sorption as well as stabilization of aggregate particles via cation bridges that induce flocculation of OM particles. While OM availability on two-stage floodplains has been documented, it is essential to understand how these OM-cation interactions change over time to better inform two-stage management practices.

To date, few studies have assessed long-term changes in soil properties on restored two-stage floodplains, particularly more than 10 years after construction or in comparison to adjacent source fields. Here, we quantify soil biogeochemical characteristics across three two-stage ditches within the same stream channel, spanning 5–16 years since construction, and compare them to adjacent agricultural fields. Because two-stage ditch construction exposes deeper subsoils (~120 cm), this chronosequence design allowed us to evaluate how soil properties evolved over time relative to their source fields from which they were excavated.

We hypothesized that two-stage floodplain soils would differ from adjacent agricultural fields and that these differences would become more pronounced with two-stage age. Specifically, we predicted that (hypothesis 1) older two-stage reaches would possess greater % OM than new reaches, potentially enhancing N retention, and (hypothesis 2a) that metal cations (Fe, Al, and Ca) would influence P retention, potentially making older floodplains more likely to act as P sinks. Furthermore, we predicted that (hypothesis 2b) variation in pH, reflecting differences in cation content, would influence the potential for P sequestration across floodplains and adjacent fields. Finally, we predicted that (hypothesis 3) tile drain flowpaths, which experience near-constant inundation, would differ from floodplains in % OM, nutrient concentrations, and cation content due to distinct hydrologic and redox conditions. Overall, we expected that observed soil patterns would reflect both the legacy of adjacent field soils and the cumulative biogeochemical processes on the two-stage floodplains, with older sections showing the greatest divergence from adjacent field conditions.

2. Methods

2.1. Field site

We conducted our study in a small agricultural watershed located in north-central Indiana, USA. The Shatto Ditch Watershed (SDW) makes up 1403 ha of the Wabash River Basin and reflects a typical midwestern agricultural watershed: over 85% of the SDW land area is planted with corn-soybean row crop rotations and is managed by multiple agricultural producers/landowners; additionally, there is one poultry concentrated animal feeding operation (CAFO) in the watershed (Trentman et al., 2021).

The soils in the SDW are patchy and diverse, but the dominant soil types are Alfisols (77%) and Mollisols (13%) (Speir et al., 2021) with runoff potential varying from “low” to “moderately high” depending on the soil classification. More than 79% of the agricultural land in the SDW is drained by subsurface tile drainage that transports excess water into the adjacent ditch which is eventually transported to the Tippecanoe River (Gökkaya et al., 2017). Soil tillage in the SDW varies, but most fields are subjected to varying degrees of conservation-tillage or no-till strategies. For example, in the 2016 water year (WY), 60% of agricultural fields sampled in the SDW used a variety of no- or conservation-tillage systems (Christopher et al., 2021). Additionally, the use of winter cover crops in the SDW varies from year to year, with an average of 62% of the agricultural land in the SDW being planted with cover crops between 2013 and 2023. Cereal rye and annual rye are the most common cover crops used in the SDW, and they are chemically or mechanically terminated approximately two weeks before planting of cash crops in the spring (Christopher et al., 2017). Inorganic N fertilizers are applied to most of the SDW agricultural fields in the spring and as a side-dressing in early summer. Inorganic P fertilizer use is rare due to relatively high levels of legacy P from historical chicken litter application from the CAFO; intermittent P application may occur based on decisions informed by soil test P (STP) results on a field-by-field basis.

The two-stage floodplain sections assessed in our study were constructed in 2007, 2017, and 2018, and are 0.8, 3.7, and 2.7 km long, respectively (Fig. 1C). The floodplain dimensions were based on calculations to determine the appropriate floodplain depth and width to stabilize banks, maintain flow in the main channel during base flow, and inundate benches 10–12 times per year (D’Ambrosio et al., 2015; Mahl et al., 2015). In the 2007 section, the average floodplain width is 3 m on each side of the main channel, which increases the bioreactive surface area 3x during inundation. In contrast, for the 2017 and 2018 sections, the average floodplain width is 1.5 m on each side of the main channel, which doubles the bioreactive surface area during inundation (Speir et al., 2020). Two-stage floodplains inundate approximately 91 days each year (Speir et al., 2020). There are also 22 tiles that drain fields throughout the watershed; of these we identified six drains which empty directly onto the floodplains during base flow conditions, forming a flowpath before the tile water enters the stream. We collected soils across transects of the floodplains adjacent to these six drains to represent physical and chemical connections between floodplains, fields, and tile flowpaths. We note that reed canary grass (*Phalaris arundinacea*) and rice cut grass (*Leersia oryzoides*) dominated floodplain vegetation in all three reaches (Roley et al., 2014), and vegetation density was also comparable among reaches at the time of sampling.

2.2. Soil sampling

For this study, we collected all two-stage soil samples on July 21, 2023. Using standard soil sampling methods, we collected soil from the top 5 cm using Oakfield soil samplers. A “wet tip” with an interior diameter of approximately 1.9 cm was used to collect soil samples from the tile drain flowpaths, while a “regular tip” with the same interior diameter was used to collect samples from the part of the floodplain that is not underneath tile flow and is only inundated during high flow events. These two soil “environments” are hereinafter referred to as the “tile flowpath” and “floodplain” soil samples, respectively.

Our study was part of a long-term project in the SDW, where a two-stage ditch was constructed in three phases over 12 years. We sampled sites adjacent to every tile drain that emptied onto floodplains in each two-stage reach throughout the watershed. Due to landowner consent and tile placement, we were constrained with an unbalanced design and limited replication. We collected soil samples from a total of six floodplain sites spanning the three two-stage reach ages. Our sites were located as follows: one floodplain site in the 2007 reach, two in the 2017 reach, and three in the 2018 reach (Fig. 1C). At the time of sampling, these two-stage sections were 16, 6, and 5 years old, respectively. On each floodplain within a site, we collected soil samples in a total of four parallel transects spaced 5 m apart from one another to capture variation along each reach (Fig. 1B). Each transect was oriented perpendicular to the stream and composed of 10 pooled and homogenized soil cores collected every 0.5 m along the length of the transect. The first transect at each site was collected from the tile flowpath between the tile drain outlet and stream, while the other three transects were collected from the two-stage floodplain. We transported all samples from the field to the laboratory in polyethylene Ziploc® bags on ice to be immediately prepped for laboratory analyses.

From Autumn 2013 through summer 2023, we collected agricultural field soil samples every Autumn (post-harvest) and Spring (before cover crop termination) in the SDW. These soil samples were similarly collected using Oakfield soil samplers at three transects along a line perpendicular to each field edge, at three sampling transects which were

40 m, 70 m, and 100 m away from the edge. At each location, six soil cores were collected from each of the 0–5, 5–20, 20–40, and 40–50 cm depths. Cores were pooled by their depths to create three separate soil samples from each field at each depth profile (Christopher et al., 2021). We only included data from the fields adjacent to the tile drain outlets included in our study. These field samples were used for two distinct purposes and grouped accordingly. First, we compiled 0–5 cm field data collected during the same WY as our two-stage sampling (i.e., 2023). We further subdivided this data by the corresponding tile outlets that they drain into to establish reference characteristics against which to compare to the two-stage soil data. Secondly, we grouped field data from all available water years by depth profile to characterize variability in soil characteristics across depth in this region, since the two-stage ditches themselves are set approximately 120 cm below field level (Fig. 1A) (Frankenberger et al., 2014).

2.3. Soil laboratory analyses

Upon return to the lab, we air dried (in aluminum pie tins), sieved (2 mm), and ground all soil samples in a Wiley Mill prior to laboratory analyses. We stored the dried samples in sealed polyethylene Ziploc® bags at room temperature in dark conditions. We then sent soil samples to Brookside Laboratories, Inc. (New Breman, OH) for standard soil testing. Briefly, analyses were performed to quantify pH (soil:water = 1:1) (Mclean, 1983) and a Mehlich III soil test for P (hereinafter referred to as Soil Test Phosphorus, STP), Al, Fe, K, Ca, Mg, Mn, Zn, and Na (Mehlich, 1984). Brookside Laboratories also measured loss on ignition % OM by calculating the relative mass difference between each soil sample when desiccated at 110°C and when baked at 360°C for 2 h in a muffle furnace (Schulte and Hopkins, 1996).

To estimate source/sink dynamics, we first calculated soil phosphorus saturation ratio (PSR) as the molar ratio of P to the sum of Fe and Al (i.e., species that bind P in soils) using Mehlich III extraction data (Sims et al., 2002):

$$PSR = \left(\frac{\frac{Mehlich\ P}{30.974}}{\frac{Mehlich\ Fe}{55.845} + \frac{Mehlich\ Al}{26.9815}} \right)$$

We then calculated soil P sorption capacity (SPSC) for each soil sample as the difference between PSR and the threshold PSR of 0.11 for our soil type of Alfisols/Mollisols (Laboski and Lamb, 2004; Nair, 2014). We then converted SPSC units from a molar ratio (i.e., the units of PSR) to units of mg P kg^{-1} soil dry mass using the following equation (Dari et al., 2018):

$$SPSC\ (mg\ P\ kg^{-1}) = (Threshold\ PSR - PSR) * \left(\frac{Mehlich\ Fe}{55.845} + \frac{Mehlich\ Al}{26.9815} \right) * 30.974$$

2.4. Data analysis

We assessed the effect of the two-stage ditch on soil characteristics by fitting linear mixed effects models (LMMs) of the form: Soil Metric ~ Construction Year + (1|Site), where site was included as a random intercept to account for the non-independence of transects nested within sites. Kenward-Roger degrees of freedom correction was applied using the *lmerTest* package (Kuznetsova et al., 2017) to account for the unbalanced design, and pairwise post-hoc comparisons were conducted using the *emmeans* package (Lenth, 2023) with Tukey adjustment, which is statistically equivalent to Tukey's HSD but implemented within the LMM framework. To assess whether spatial position within the watershed could explain patterns attributed to restoration age, we also fit competing LMMs using the distance from the most upstream site (DFMUS) as the sole fixed predictor, retaining site as a random intercept, and compared models via AIC, BIC, and marginal R^2.

Due to insufficient sample size ($n < 3$, SI Table 2 A), we excluded the 6- and 16-year-old (hereafter "-yr") tile flowpaths from statistical analyses but still included them in data visualizations to give context to the other values. In cases when exclusion of these groups from statistical analyses resulted in the comparison of two groups of data, we used the Mann-Whitney *U* test to account for non-normal distributions of the remaining two data groups. Comparisons between SPSC and the reference value (i.e., zero) were conducted using the Wilcoxon signed-rank test for flowpath and floodplain data and the *t*-test for field data. We used a significance level of $\alpha = 0.05$ for all analyses.

We calculated rank correlations primarily using Kendall's method (coefficient τ) because the variables showed non-normal distributions. In cases where data approximated normal distributions, we used the Pearson method (correlation coefficient r). Finally, we used the R language version 4.3.1 (R Core Team, 2023) to run all statistical analyses and the *ggplot2* (Wickham, 2020) and *cowplot* (Wilke, 2020) packages to create all figures.

3. Results

3.1. OM accumulates with floodplain age

Soil % OM trends differed based on the different environments. Specifically, % OM was comparable amongst the fields adjacent to the 5-, 6-, and 16-yr two-stage reaches ($p > 0.05$; Fig. 2A). Despite a lack of significant differences among fields from which two-stage ditches were constructed, % OM was ~3x higher for the 16-yr floodplain compared to the more recently constructed floodplains ($p < 0.05$; Fig. 2A). Furthermore, % OM in the 16-yr floodplain was > 3x higher compared to the adjacent field ($p < 0.05$; SI Figure 3 A). However, for the 5- and 6-yr reaches, % OM did not increase on floodplains relative to the adjacent field (SI Figure 3 A). Of note, the variation in floodplain % OM was better explained by the two-stage ditch age as compared to the geographic orientation of the two-stage ditch as defined by the DFMUS (AIC 21.8 vs. 38.9; BIC 25.6 vs. 42.0; marginal R^2 0.98 vs. 0.55, respectively).

We also compared soil samples collected from the floodplains (which are only inundated during high flows) to samples collected from the tile flowpaths, which are: (1) inundated with tile water for most of the year, and (2) incised due to erosion by tile water. We found that mean % OM was approximately 4x higher for the 16-yr floodplain sites compared to the mean of available tile flowpath samples (SI Figure 3 A); however, insufficient flowpath replication ($n < 3$) precluded formal statistical analysis of this relationship (SI Table 2 A).

By comparing soil characteristics in the excavated two-stage floodplain to those in adjacent, unexcavated agricultural fields across depth profiles, we can assess how two-stage implementation has altered soils at their new surface depths relative to the conditions that would have existed prior to excavation. As expected, % OM decreased with depth ($p < 0.05$; SI Table 1). For the 5- and 6-yr reaches, tile flowpat% OM was similar to % OM at the 40–50 cm depth for fields (Fig. 2A). In contrast, floodplain % OM was 2–3x higher than % OM for the deepest field soil samples (Fig. 2A).

3.2. NO_3^--N & NH_4^+-N accumulate on older floodplains

Soil NO_3^--N in the fields had a mean ± SE of 5.76 ± 0.69 mg kg^{-1} (range: 1.3–22.8 mg kg^{-1}) and was lowest in the field adjacent to the 16-yr two-stage reach, although this trend was not statistically significant ($p > 0.05$; Fig. 2B. In contrast, soil NO_3^--N on floodplains had a mean ± SE of 2.13 ± 0.55 mg kg^{-1} (ranged from 1.3 to 8.0 mg kg^{-1}) and was highest in the 16-yr floodplains ($p < 0.05$; Fig. 2B). Soil NO_3^--N in the tile drain flowpath trended slightly higher than for the floodplain in the two newer reaches but was 10x lower than the floodplain in the 16-yr reach (SI Figure 3 H). Of note, the variation in

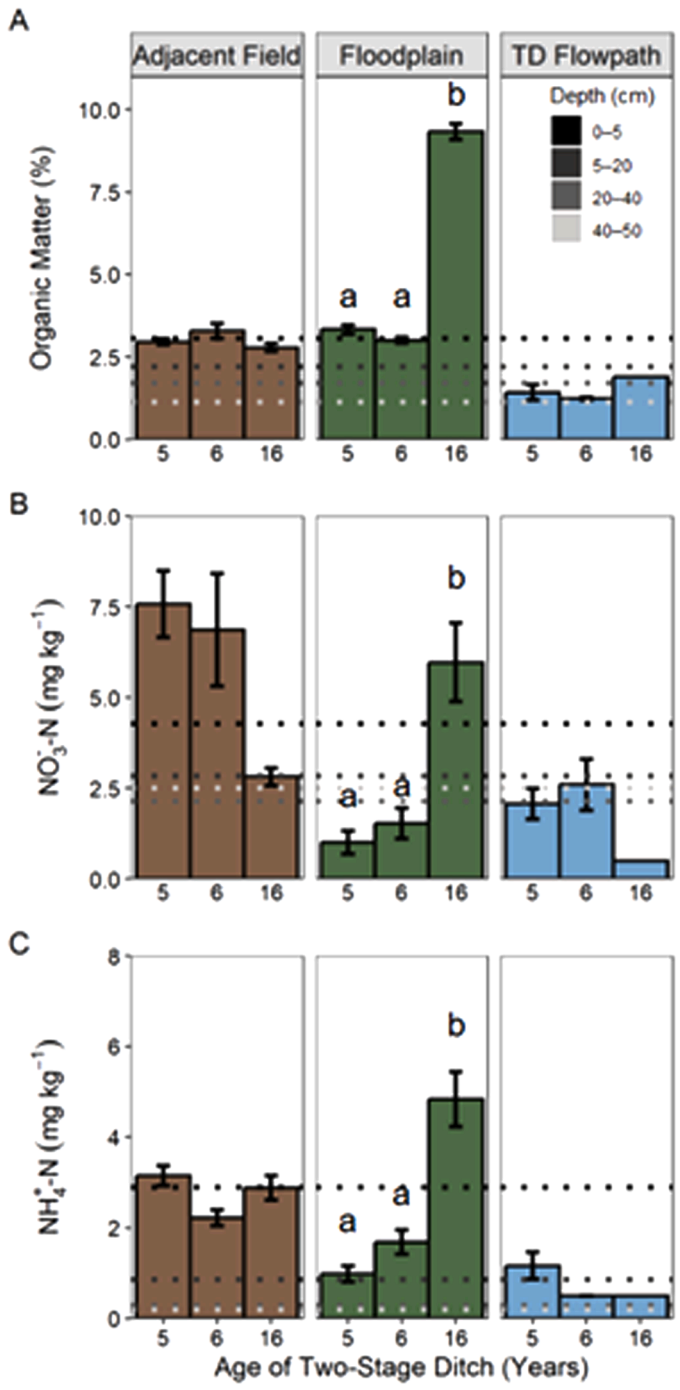


Fig. 2. Soil characteristics grouped by environment type. Mean ± SE values for **A)** Organic Matter **B)** NO_3^--N, and **C)** NH_4^+-N for soils from 0 to 5 cm, subdivided by environment type and plotted for the age of the nearest two-stage reach. The dashed lines correspond to the soil characteristics' mean values across depth profiles from the fields adjacent to the tile drains, sampled semiannually from 2013 to 2023 to serve as a reference for the region's soil characteristics (SI Table 1). Note: n < 3 for 6- and 16-yr tile flowpaths (SI Table 2).

floodplain NO_3^--N was moderately better explained by the two-stage ditch age as compared to the geographic orientation of the two-stage ditch as defined by the DFMUS (AIC 55.0 vs. 62.2; BIC 58.8 vs. 65.3; marginal R^2 0.74 vs. 0.59, respectively).

Soil NH_4^+-N on the fields had a mean ± SE of 2.75 ± 0.14 mg kg^{-1} (range: 1.3–4.3 mg kg^{-1}) on fields adjacent to the two-stage. Similar to NO_3^--N, floodplain soil NH_4^+-N trended upwards with two-stage age ($p < 0.05$; Fig. 2C). Soil NH_4^+-N in the tile flowpath tended to be lower than the floodplain in the older reaches (SI Figure 3I). Similar to NO_3^--N, NHO_4^+-N on the floodplain was moderately better predicted by age than by DFMUS (AIC 40.9 vs. 46.9; BIC 44.8 vs. 50.0; marginal R^2 0.82 vs. 0.72, respectively).

3.3. Floodplains show moderate STP retention relative to fields

Although we did not observe significant differences among fields for STP ($p > 0.05$; Fig. 3B), we noted that the field adjacent to the 16-yr reach had a higher mean STP than the other fields. We found that STP was generally higher in adjacent fields than the floodplains, although this difference was only statistically significant in the 5-yr reach ($p < 0.05$; SI Figure 3B). Similar to % OM, STP was lower in the tile flowpath compared to the floodplain in the 6- and 16-yr reaches (although statistical analysis was not possible in the 16-yr reach and was non-significant in the 6-yr reach; SI Figure 3B). We also found that STP in the fields decreased with depth ($p < 0.05$; SI Table 1), and that STP for the 40–50 cm depth in the fields was similar to STP for samples taken from the tile flowpath in all reaches (Fig. 3B). Amongst the floodplain data, only the 5-yr two-stage reach had similar STP compared to the 40–50 cm field STP (Fig. 3B). For the 6- and 16-yr reaches, floodplain STP was similar to the 20–40 cm depth on the fields (Fig. 3B). Although neither were great models (which aligns with a lack of statistical significance in Fig. 3B), the variation in floodplain STP was better explained by the two-stage ditch age as compared to the geographic orientation of the two-stage ditch as defined by the DFMUS (AIC 134 vs. 145; BIC 138 vs. 148; marginal R^2 0.27 vs. 0.20, respectively).

3.4. Two-stage floodplains concentrate Fe and Ca while surface Al drops

We observed a unique but non-significant pattern in Mehlich Al among fields, with the field adjacent to the 6-yr reach having the lowest mean Mehlich Al (Fig. 3C). Soils from floodplains mirrored this pattern (Fig. 3C). On average, adjacent fields had higher Mehlich Al than the floodplains, which had higher Mehlich Al than the tile flowpaths ($p < 0.05$; SI Figure 3D). However, depth data from the adjacent fields indicated that Mehlich Al trended upwards as depth increased in the field (SI Table 1). Mehlich Al on floodplains was poorly predicted by both two-stage age and DFMUS (AIC 171 vs. 185; BIC 175 vs. 188; marginal R^2 0.24 vs. 0.09, respectively). However, the floodplain soil Al, exposed by excavation, was much lower than expected based on the relatively high Mehlich Al found in the deeper field soils (Fig. 3C).

We observed a pattern shared by the fields and floodplains in which Mehlich Fe was higher on the 16-yr reach than the 5-yr reach ($p < 0.05$; Fig. 3D). Mehlich Fe was lowest at the deepest field profile (SI Table 1) and was higher on floodplains than adjacent fields across all two-stage ages ($p < 0.05$; SI Figure 3E). However, we did not observe any consistent patterns when comparing the tile flowpath to the floodplain (SI Figure 3E). The variation in floodplain Mehlich Fe was poorly explained by both the two-stage ditch age as compared to the geographic orientation of the two-stage ditch as defined by the DFMUS, although age formed a moderately better model (AIC 152 vs. 166; BIC 155 vs. 169; marginal R^2 0.78 vs. 0.72, respectively).

We found that Mehlich Ca on the fields adjacent to the 6-yr reach was significantly higher than the fields adjacent to the 16-yr reach ($p < 0.05$; SI Figure 1); however, this pattern did not persist for the floodplain soils (SI Figure 1). Mehlich Ca trends between field and floodplain mirrored those of Mehlich Fe, with two-stage floodplains having higher Mehlich Ca than adjacent fields for all two-stage age groups ($p < 0.05$; SI Figure 3 F). Tile flowpath Mehlich Ca trended higher than floodplain Mehlich Ca for the 5- and 6-yr reaches but lower than floodplain Mehlich Ca for the 16-yr reach although statistical comparison was only possible in the 5-yr reach ($p < 0.05$; SI Figure 3 F). Amongst field soils, we found that Mehlich Ca concentrations gradually declined as depth increased although this trend was non-significant ($p > 0.05$; SI Table 1). Finally, floodplain Mehlich Ca at all reaches was ~2x higher than Mehlich Ca at

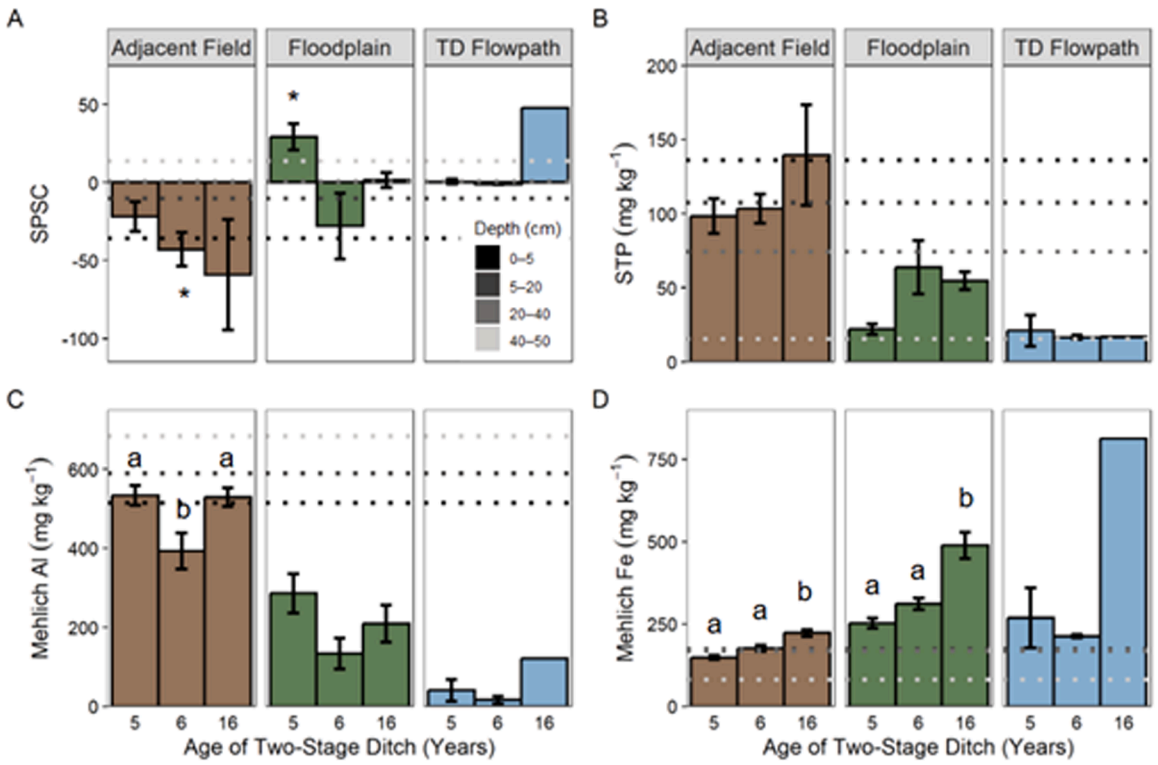


Fig. 3. Soil characteristics grouped by environment type. Mean ± SE values for **A)** SPSC, **B)** STP, **C)** Mehlich Al, and **D)** Mehlich Fe for soils from 0 to 5 cm, subdivided by environment type and plotted for the age of the nearest two-stage reach. The dashed lines correspond to the soil characteristics' mean values across depth profiles from the fields adjacent to the tile drains, sampled semiannually from 2013 to 2023 to serve as a reference for the region's soil characteristics (SI Table 1). Stars (*) in panel **A** indicate SPSC values significantly different from 0. Calculated SPSC values above 0 indicate soils that are likely to be P sinks, while calculated SPSC values below 0 indicate soils that are likely to be P sources. Note: n < 3 for 6- and 16-yr tile flowpaths (SI Table 2).

the deepest field soil profiles (SI Figure 1). The variation in floodplain Ca was better explained by two-stage ditch age than DFMUS, although neither model was particularly strong (AIC 223 vs. 238; BIC 226 vs. 242; marginal R^2 0.10 vs. 0.13, respectively).

3.5. Two-stage floodplains reduce P source potential in soils

When we calculated SPSC values to estimate whether soils are likely to be a P source (SPSC < 0) or a P sink (SPSC > 0), we found that all field averages trended towards being a P source, although only the 5- and 6-yr reaches' SPSC values were significantly lower than 0 ($p < 0.05$; Fig. 3A). In contrast, floodplain SPSC values differed among restoration age classes, and only the 5-yr floodplain was significantly greater than 0 ($p < 0.05$; Fig. 3A), suggesting that the youngest floodplain acted as a P sink while the 6- and 16-yr floodplains were likely neither net P sinks nor net P sources. Linear models predicting floodplain SPSC based on age of the two-stage ditch were both poor, although the age model was slightly more accurate than DFMUS (AIC 138 vs. 150; BIC 142 vs. 153; marginal R^2 0.29 vs. 0.12, respectively). Field SPSC values were generally more negative compared to the adjacent floodplains' and flowpaths' SPSC values (SI Figure 3 C); SPSC values for flowpaths on the two newer two-stage reaches suggest that the soils were neither a P source nor a P sink (i.e., SPSC did not differ from 0; $p > 0.05$; Fig. 3A), while SPSC for the flowpath on the oldest floodplain indicated that soils were likely a P source (SPSC > 0; n = 1; Fig. 3A). We also note that SPSC for fields became more positive with depth with the top 20 cm of soil likely being a P sink (SPSC < 0; $p < 0.05$), soil 20–40 cm deep likely being neither a sink nor a source (SPSC did not differ from 0; $p > 0.05$), and soil 40–50 cm deep being more likely to be a source (SPSC > 0; $p < 0.05$) (SI Table 1).

3.6. Two-stage floodplains and flowpaths maintain higher pH than fields

We found that pH differed among the fields, although this trend was not statistically significant ($p > 0.05$; SI Figure 2). Similarly, we did not observe any significant differences among the two-stage reaches' floodplains and flowpaths; furthermore, pH for floodplains (ranging from 7.5 to 8.0) was similar to pH in tile flowpaths (ranging from 7.8 to 8.1) for all three two-stage age groups (SI Figure 3 G). Although pH was consistently higher on two-stage floodplains and flowpaths compared to fields (SI Figure 3 G), field pH did not significantly vary with depth ($p > 0.05$; SI Table 1). Of note, the variation in floodplain pH was similarly explained by models based on two-stage ditch age as well as based on DFMUS (AIC −7.7 vs. −7.6; BIC −3.8 vs. −4.5; marginal R^2 −0.15 vs. 0.001, respectively).

3.7. Cation and pH correlations highlight potential factors associated with P retention

When we performed correlations to test for effects of floodplain soil % OM accumulation on other soil properties, we found a weak positive correlation with NO_3^--N ($\tau = 0.238$; $p < 0.05$) but we did not observe significant relationships with SPSC, STP, cations, or pH. When we compared STP to variables known to influence P cycling, we found a strong, negative correlation between SPSC and STP as expected ($\tau = -0.645$; $p < 0.05$; SI Figure 4B). We also found that pH was strongly negatively correlated with Mehlich Al ($\tau = -0.719$; $p < 0.05$; Fig. 4A) across all environments combined. When analyzed separately by location, this negative correlation between pH and Mehlich Al was consistent across all three environment types: tile flowpath ($\tau = -0.416$; $p = 0.04$), field ($r = -0.806$; $p < 0.001$), and floodplain ($\tau = -0.856$; $p = 0.02$). Although direct statistical comparison across environment types is precluded by the use of different correlation methods (Kendall's

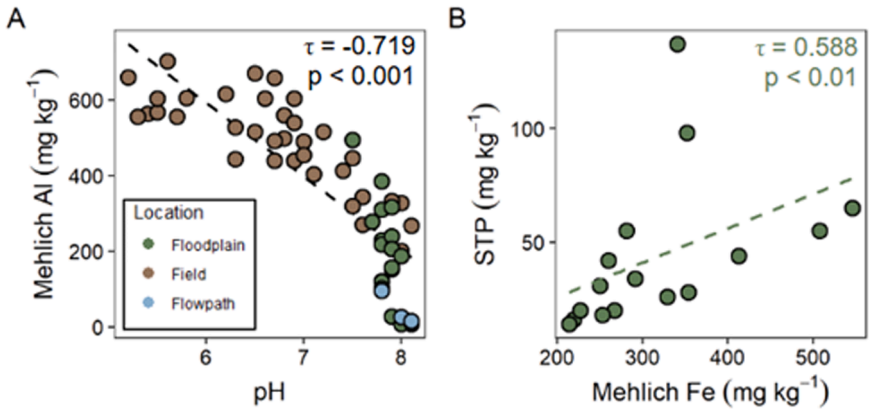


Fig. 4. Scatterplots of pH vs. Mehlich Al and Mehlich Fe vs. STP. A.) Scatterplot comparing Mehlich Al and pH across floodplain, field, and flowpath. The dashed line is black to represent that the correlation accounts for all locations combined together. B.) Scatterplot comparing STP and Mehlich Fe in the floodplain. The dashed line is green to represent that the correlation accounts only for the floodplain.

τ vs. Pearson's r), the floodplain exhibited the strongest association between pH and Mehlich Al. We also noted greater pH variation in the field data in comparison to the floodplain and flowpath data (Fig. 4A). Furthermore, two-stage floodplain STP was positively correlated with floodplain Mehlich Fe ($\tau = 0.588$; $p < 0.05$; Fig. 4B) and floodplain MehlichAl + Fe ($\tau = 0.750$; $p < 0.05$; SI Figure 5).

4. Discussion

4.1. Organic matter builds over time, potentially enhancing denitrification potential

Our data suggests that two-stage age explains more variation in floodplain OM than does distance from the most upstream site (AIC 21.8 vs. 38.9; BIC 25.6 vs. 42.0; marginal R^2 0.98 vs. 0.55, respectively), although we acknowledge that spatial positioning and two-stage age are not fully independent in our study design. As such, our data aligns with previous studies suggesting that OM can accumulate gradually on two-stage floodplains over time (Fig. 2A) (Hanrahan et al., 2018; Mahl et al., 2015) likely driven by cycles of vegetative growth and floodplain sediment deposition (Burchell et al., 2007; Zhai et al., 2013). Given that % OM typically decreases as soil depth increases (SI Table 1) (Kedir et al., 2022; Yang et al., 2016; Yu et al., 2019), we expect % OM on recently excavated two-stages to be relatively low, which could impede biogeochemical processes (e.g., denitrification) on recently constructed floodplains (Speir et al., 2020). For the 5- and 6-year-old reaches, we found that floodplain % OM was ~3x higher than field % OM at a depth of 40–50 cm, which is approximately 1/3–1/2 the full depth of the floodplain (120 cm), and expected to have greater % OM than soil at the full depth of the floodplain (Fig. 2A). Furthermore, % OM was ~3x higher on the 16-year-old floodplains compared to the newer floodplains (Fig. 2A), which (in combination with AIC-backed support of the age-based model's superiority over the DFMUS-based model) suggests that OM accumulation has continued for over a decade without any two-stage maintenance. Lower % OM in the tile flowpaths compared to floodplains is likely a combined result of reduced plant growth caused by anoxic stress in the constantly-inundated tile flowpaths (Barclay and Crawford, 1982) and the scouring of sediment by the water that runs through them. However, OM accumulation in the tile flowpaths is possible when flow rates are low (e.g., summer drought or winter frozen conditions) as that is when plant matter may accumulate and eventually decompose.

Decomposition rates of soil OM in two-stage ditches were found to be comparable to those in traditional trapezoidal ditches (Griffiths et al., 2012), suggesting that elevated OM on two-stage floodplains represents a stable, long-term pool rather than a transient one. Furthermore, increased OM may contribute to higher denitrification rates on restored floodplains, consistent with previous findings conducted in the SDW (Hanrahan et al., 2018; Roley et al., 2012b; Speir et al., 2020). Increased OM is especially important during times of inundation, when OM (rather than NO_3^--N) becomes the limiting factor in denitrification on two-stage floodplains (Hanrahan et al., 2018; Roley et al., 2012b; Speir et al., 2020). An increase in OM across the entire two-stage reach would thus provide increased denitrification potential during inundation despite the fact that inundation patterns vary along the reach due to hydrologic, geologic, and agricultural differences (Mahl et al., 2015).

We did not find a significant correlation between % OM and Mehlich Al despite previous observations of a negative correlation between OM and Al (Li et al., 2022), which is likely due to the fact that the Mehlich III soil test measures all forms of Al, be they soluble or insoluble, while Li et al. (2022) only measured soluble Al.

4.2. Nitrogen patterns on two-stage floodplains reflect hydrology and OM

Average soil NO_3^--N was higher in the fields than on the floodplains for the newer two-stage reaches while the opposite was true for the 16-yr reach (SI Figure 3 H). Given that NO_3^--N is a highly leachable nutrient (Gaines and Gaines, 1994) and is unlikely to remain in high quantities long after agricultural fertilizer application, it is likely that the low NO_3^--N concentration in field soils adjacent to the 16-yr two-stage reach is a result of differences in fertilizer application rather than local variation. It is likely that N fertilizer had not yet been applied to the field adjacent to the 16-yr reach at the time of the Spring 2023 sampling whereas fertilizer had likely already been applied to the fields adjacent to the newer two-stage ditches.

While the soil NO_3^--N differences in the fields may be explained by fertilizer application (or lack thereof) the differences in the floodplains were likely due in part to the orientation of the watershed. NO_3^--N was elevated on the 16-yr floodplain relative to younger floodplains; while downstream loading and denitrification have been proposed as opposing controls on floodplain NO_3^--N in our study system (see Speir et al., 2020 and Speir et al., 2022), the net outcome cannot be determined from our observational data alone. However, our linear models for prediction of NO_3^--N on floodplain soils are moderately better predicted by two-stage age than by DFMUS (AIC 55.0 vs. 62.2; BIC 58.8 vs. 65.3; marginal R^2 0.74 vs. 0.59, respectively), suggesting that elevated NO_3^--N on the 16-yr floodplain relative to younger floodplains exceeded what would be expected based on field depth alone (Fig. 2B).

Given that microbially-mediated denitrification requires OM as a C source (Hanrahan et al., 2018; Knowles, 1982; Starr and Gillham, 1993), the high % OM on the 16-yr floodplain may contribute to greater potential for NO_3^--N removal from the system compared to the younger

floodplains, which have lower % OM (Fig. 2A). However, denitrification rates have been shown to plateau as NO_3^--N loads increase (Roley et al., 2012a), thereby limiting the beneficial effects of the high % OM, and highlighting a need to reduce NO_3^--N runoff via additional stacked-management practices.

The floodplain in the 5-yr reach had lower NH_4^+-N than did the adjacent fields, but this trend was reversed in the 16-yr reach and there was no difference for the 6-yr reach (SI Figure 3I). However, differences in local NH_4^+-N concentrations as suggested by differences in field NH_4^+-N (Fig. 2C), although non-significant, imply that these comparisons may not accurately reflect changing NH_4^+-N concentrations in two-stage ditches over time. However, there may be accumulation of floodplain NH_4^+-N over time, with the 16-yr floodplain having higher concentrations of NH_4^+-N than the younger floodplains; furthermore, all of the floodplains have higher NH_4^+-N than depth alone would indicate (Fig. 2C). This high concentration of NH_4^+-N on the 16-yr floodplain was plausibly due to the mineralization of higher OM on the floodplain (Grzyb et al., 2020).

4.3. *Two-stage floodplains: complex factors associated with soil P sink potential*

Decreased STP on two-stage floodplains and tile flowpaths relative to the adjacent fields (SI Figure 3B) was likely a result of natural decreases in STP as soil depth increases (SI Table 1). This conclusion corroborates prior findings that P in agricultural systems with historic manure applications exhibits an inverse relationship with soil depth due to extraction of P as the nutrient percolates through the soil layers (Kedir et al., 2022; Torbert et al., 2002). However, local STP trends across soil depth would predict lower concentrations of STP than are observed on the floodplains and potentially the tile flowpaths as well, suggestive of P accumulation in two-stage systems due to processes occurring within the system itself (Fig. 3B). This indication of STP accumulation suggests that removing the top layer of soil during two-stage floodplain construction did not result in continuous leaching of STP into the streams.

We note that application of soil P amendments is rare in the SDW. As such, the accumulation of STP in two-stage floodplains is not merely a result of short-term runoff from adjacent fields. Additionally, STP is likely not accumulating from OM mineralization, as there was no correlation found between STP and % OM (SI Figure 4 A); rather, it likely exists as residual P from historical chicken litter application (from the collocated CAFO) to the soil from which the two-stage ditches were excavated.

There are many factors contributing to P retention in soils, such as pH, ionic strength, ion identity, soil type, and concentrations of ions that may influence P binding (Arai and Sparks, 2007; Essington, 2015; Fan et al., 2020; Gustafsson et al., 2012; Wan et al., 2019; Yi et al., 2023). For example, Al and Fe are known to be two important cations responsible for soil P complexation (Penn and Camberato, 2019). We found that Mehlich Al did not show significant accumulation within the two-stage reaches (Fig. 3C), although Mehlich Fe in the floodplain and tile flowpath were present at levels above those predicted by the trends of field depth data (Fig. 3D) as well as the levels present on the adjacent fields during the time of this study (SI Figure 3E). Mehlich Al results did not appear to be influenced by age, however, as similar Mehlich Al patterns were observed across ages in all three environment types (Fig. 3C) and linear modelling indicated both age and DFMUS as poor predictors of floodplain Mehlich Al (AIC 171 vs. 185; BIC 175 vs. 188; marginal R^2 0.24 vs. 0.09, respectively).

The subsurface tile drain system in agricultural fields maintains efficient soil drainage and aeration. Conversely, tile flowpaths experience near-constant inundation while floodplains are periodically inundated, meaning neither are well-aerated throughout the year. As such, two-stage ditches are more likely to experience anaerobic conditions than are the adjacent agricultural fields (Speir et al., 2020). In these anaerobic conditions in two-stage reaches, Fe(III) is likely reduced to Fe (II), causing a release of soil bound P, while aerobic fields do not experience such reduction of Fe(III) (Schmidt, 1999). We observed a positive correlation between STP and Mehlich Fe in floodplain soils (Fig. 4B), suggesting that Fe may contribute to P sequestration on two-stage floodplains which is in alignment with known interactions between Fe and P (Penn and Camberato, 2019). However, because Mehlich III selectively extracts Fe(II) rather than Fe(III), comparisons of Fe between the anaerobic floodplain and aerobic field environments are confounded by differences in Fe speciation and are therefore not discussed further.

Within the pH range of ~5 to ~8 that is observed in the WY 2023 dataset, Mehlich Al concentrations were highest at low pH and decreased as pH increased (Fig. 4A). This elevated Mehlich Al concentration at pH 5–6 corresponds to the approximate pH at which Al maximally binds soil P (SI Figure 6), plausibly contributing to increased retention of P by Al (Penn and Camberato, 2019). In general, the chemical processes contributing to maximal P retention by Al as pH decreases are the result of two main reactions occurring within soils and which are summarized by the net reaction $Al(OH)_{3(s)} + H^+ + H_2PO_{4(aq)}^- \rightleftharpoons AlPO_4 \cdot 2\,H_2O_{(s)} + H_2O$ (Mulder and Stein, 1994; Penn and Camberato, 2019). As pH decreases, H^+ increases and Le Chatelier's principle dictates that the net reaction equilibrium lies to the right, pulling aqueous $H_2PO_4^-$ out of solution and into a solid form bound by Al. Although low pH permits increased P sequestration, thereby preventing contamination of local waterways, the high concentration of aqueous Al that is associated with low pH may also contribute to plant toxicity (Keltjens and Van Loenen, 1989).

Although Mehlich Fe was higher on the floodplains than in the fields (SI Figure 3E), this trend was opposite of that for Mehlich Al, which was higher in field soils than on floodplains (SI Figure 3D). Inconsistencies we found in Mehlich Al and Mehlich Fe trends suggest that SPSC is governed primarily by changes in STP rather than changes in MehlichAl + Fe (SI Figure 4B; SI Figure 5 A). The opposing trends in Mehlich Al and Mehlich Fe between floodplain and field soils likely reflect two distinct processes. Lower Mehlich Al on floodplains relative to adjacent fields is consistent with the excavation of Al-depleted surface soils during two-stage construction, given that Mehlich Al increased with depth in intact field profiles, suggesting that the material exposed at the floodplain surface was originally Al-poor subsurface material. Conversely, higher Mehlich Fe on floodplains likely reflects periodic anaerobic conditions during inundation, which promote reduction of Fe(III) to Fe (II), the form measured by Mehlich III. Because these two processes drive Mehlich Al and Mehlich Fe in opposing directions, their combined effect on P sorption capacity is partially self-canceling, leaving STP as the primary source of variation in SPSC across environments.

Importantly, SPSC and its components' trends were variable on the two-stage floodplain (Fig. 3); in fact, the 5-, 6-, and 16-yr floodplains each exhibited a different relative likelihood of being a P sink (although only the 5-yr reach was statistically different from 0 while the 6-yr reach was not and the 16-yr reach could not be statistically assessed, although its mean value was close to 0) (mean ± SE, 1.56 ± 4.78; Fig. 3A). Despite these findings on the two-stage floodplain, previous research has suggested that the relatively young two-stage ditches in this region are likely to be P sinks (Trentman et al., 2020). Our data indicate that although there was no consistent pattern in SPSC across floodplain age, field SPSC across depths suggest increased SPSC (i.e., greater tendency to be a P sink) as depth increased (SI Table 1).

In contrast, SPSC data suggest that the fields adjacent to the two-stage reaches are all likely to be P sources, although not all are statistically significant (Fig. 3A), likely due to residual P from historical chicken litter application. From a practical agricultural perspective, fields must be a P source for plants to access the P needed for growth, although the ecological implications of this can lead to downstream eutrophication, harmful algal blooms, and hypoxic conditions that degrade aquatic habitat quality (Chen et al., 2016; Rast and Thornton, 1996). Although SPSC is a common method used to estimate the

likelihood that a soil will be a P sink or source (Dari et al., 2018; Kedir et al., 2022; Nair, 2014), the method does not account for the binding capabilities of Ca, another cation that is responsible for P complexation in soils (Penn and Camberato, 2019). Floodplains from all three and the flowpath from two of the three floodplain age groups indicate potential Mehlich Ca accumulation compared to adjacent fields (SI Figure 3 F), and all the floodplains and tile flowpaths experienced Ca accumulation based on Mehlich Ca trends across the depth profile (SI Figure 1).

The consistently higher pH on two-stage floodplains (~pH 8) compared to adjacent fields (~6.0–7.5) approaches conditions favorable for Ca-mediated P complexation (Penn and Camberato, 2019; SI Figure 6); however, without P fractionation data we cannot confirm whether Ca-mediated retention is occurring in our system. As such, although the SPSC data do not support categorization of two-stage ditches in our system as likely P sinks, increased concentrations of Mehlich Ca, as well as idealized conditions, suggest that Ca may contribute to P sequestration before P reaches the stream. This variation in the factors that contribute to P sequestration in two-stage ditches (i.e., Al, Fe, and Ca) supports previous research concluding that although two-stage ditches may be insufficient by themselves to remove all dissolved N and P runoff, they can contribute to cleaner waterways and the eventual achievement of the goal set by the US and Canada's Great Lakes Water Quality Agreement (GLWQA) of 2012, which aims to reduce P loading by 40% (Christopher et al., 2021; Davis et al., 2015).

4.4. Two-stage ditches in practice

Our findings support the implementation of two-stage ditches within tile-drained agricultural systems, particularly those with low-gradient channelized waterways characteristic of the Midwest; however, transferability to other regions will likely depend on local hydrology, soil type, and land management context. Our findings suggest increases in soil OM over a period of 6–16 years. This increase in soil OM (hypothesis 1) not only enhances overall soil health but also provides a potential mechanism for denitrifying contaminated agricultural effluent, thereby contributing to cleaner waterways and supporting goals like those set by the GLWQA and the United States Gulf Hypoxia Action Plan (US EPA, 2015). In addition to improving soil OM, our analysis of soil chemistry within two-stage ditch floodplains indicates a capacity for P sequestration. Although the exact mechanism of this sequestration remains unclear, we hypothesize that it may involve cation-mediated processes (hypothesis 2a) coupled with pH variation (hypothesis 2b). Furthermore, while the mediation of NO_3^--N and NH_4^+-N by two-stage ditches is not entirely understood, our data suggest that older two-stage ditch floodplains may exhibit higher concentrations of NH_4^+-N, potentially due to OM mineralization. Lastly, although statistical analysis was limited by replication, we compared soil chemistry in floodplains to the nearly-constantly inundated tile drain flowpaths (hypothesis 3). In short, hypothesis 1 (Increase %OM in older floodplains) was supported, hypothesis 2 (P sequestration) was partially supported with evidence that sequestration capacity shifts with restoration age, and hypothesis 3 (N mediation) was partially supported with older floodplains showing elevated $NH4^+$-N consistent with increased OM mineralization.

The relatively low maintenance requirements of two-stage ditches, compared to other stream pollution reduction methods, and their long-term compatibility with agricultural practices, may help offset the initial implementation costs. When integrated with other on-field and near-field practices, two-stage ditches represent a crucial step towards achieving cleaner waterways and sustainable agricultural landscapes.

5. Limitations

Although our study analyzed multiple ages of two-stage reaches, replication was unbalanced because only one tile drain outlet emptied onto the floodplain within the 2007 reach, precluding additional site placement within that reach. Future work would benefit from a balanced design incorporating multiple floodplain sites per reach age, ideally across independent watersheds, to better partition the effects of restoration age from site-level variability. Additionally, the 16-yr reach was hydrologically positioned downstream of the other reaches, potentially influencing soil chemistry. We accounted for this geographic constraint by comparing AIC values of LMMs based on two-stage age as well as geographic position in the watershed as defined by DFMUS. Furthermore, geographic positioning was the basis for the regional analysis via comparison of field chemistry to establish as close of a regional reference as possible. Lastly, although the connection between OM concentration and denitrification was cited from prior literature, we did not directly measure denitrification.

CRediT authorship contribution statement

Brian A. Wagner: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jennifer L. Tank:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Mitchell J. Liddick:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ursula H. Mahl:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

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Declaration of Generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Claude to check for grammatical correctness. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.soilad.2026.100117.

Data availability

Research Link Provided

Compiled data set for: Sixteen Years of Two-Stage Floodplain Development: Long-Term Changes in Soil Chemistry and Nutrient Retention (github)

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